

Geetha Mittapalli
GIS Data Analyst
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Professional Summary

Highly skilled GIS & Data Analyst with 3+ years of experience in geospatial data management, data analytics, automation, and visualization. Expert in designing end-to-end GIS workflows, performing advanced spatial analysis, integrating large datasets, and developing interactive dashboards for decision-making. Strong technical proficiency across ArcGIS platforms, Python, SQL, cloud services, and BI tools. Proven ability to optimize data pipelines, maintain data accuracy, automate processes, and deliver scalable geospatial solutions for infrastructure, public safety, disaster response, and business operations.

Technical Skills

- **GIS & Spatial Technologies:** ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, QGIS, Geo Server, ArcGIS REST API, Geoprocessing Tools, Raster & Vector Analysis, Spatial Data Modeling.
 - **Programming & Automation:** Python (ArcPy, Pandas, GeoPandas), SQL, JavaScript for web mapping, FME Workbench, ETL Pipeline Development, Data Cleaning Automation
 - **Data Visualization & Analytics:** Tableau, Power BI, ArcGIS Dashboards, Excel Analytics, Spatial Statistics, Predictive Modeling, Time-Series Analysis
 - **Database Management:** PostGIS, SQL Server, MySQL, Spatial Indexing, Schema Design, Data Governance, Versioned Geodatabases
 - **Cloud & Deployment:** AWS (EC2, S3, RDS, Lambda), Azure GIS Services, Cloud-based GIS Deployment, API Integration
 - **Tools & Collaboration:** Git, GitHub, Jira, Agile/Scrum, Documentation, GeoJSON/REST Integration, CI/CD (basic)
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Professional Experience

AMT Engineering | GIS Data Analyst

Jun 2024 – Present

- Designed and implemented scalable geoprocessing pipelines for large spatial datasets, improving processing efficiency by 60%.
- Built advanced spatial models for land-use planning, asset tracking, and environmental risk analysis using ArcGIS Pro and Python.
- Developed automated ETL workflows that combined GPS data, cadastral datasets, satellite imagery, and survey records into unified geodatabases.

- Performed raster analysis, terrain modeling, hydrology analysis, and network analysis for infrastructure and environmental projects.
- Published enterprise-level feature services, configured web maps, and administered GIS content in ArcGIS Enterprise.
- Implemented strict QA/QC processes, achieving over 98% accuracy across complex spatial layers.
- Created interactive dashboards with ArcGIS Online and Power BI for real-time project monitoring.

Project : Citywide Utility Network Modernization

Overview:

Modernized and integrated legacy utility data (water, sewer, stormwater) into a centralized enterprise GIS to improve asset accuracy and operational efficiency.

Key Contributions:

- Converted CAD, PDFs, and GPS field data into a unified geodatabase with consistent schema and topology.
- Built automated ETL workflows using Python (ArcPy) and FME, reducing manual processing by **70%**.
- Developed a Utility Network Model with domains, connectivity rules, and asset groups for accurate representation of 150,000+ assets.
- Performed spatial analysis (network tracing, flow modeling, risk zones) to support maintenance and planning.
- Created ArcGIS Dashboards and Field Maps for real-time field updates and inspection tracking.

Impact:

- Improved asset accuracy to **98%** through QA/QC automation.
- Reduced field verification time by **60%** with real-time dashboards.
- Enabled engineering teams to retrieve critical utility information in **seconds**, enhancing operational decision-making.

City Of Chanhassen | GIS Data Analyst

Dec 2023 – May 2024

- Conducted spatial-temporal analysis on transportation and mobility datasets to identify patterns and optimize operational efficiency.
- Extracted, cleaned, and transformed large datasets using SQL, Python, and automated workflows.
- Created heatmaps, cluster analyses, and spatial trend models to support business planning.
- Designed geospatial dashboards for leadership teams to track KPIs, reducing manual reporting time by 40%.

- Developed API integrations to sync field data, IoT sensor data, and GIS services into analytics platforms.
- Collaborated with data scientists to build predictive models using geospatial variables.

Cognizant - Data Analyst / GIS Technician***Jun 2022 – May 2022***

- Developed SQL queries and Python scripts for spatial/non-spatial analytics.
- Supported ETL transformation of large tabular datasets for GIS readiness.
- Collaborated on advanced statistical modeling and BI reporting.
- Built dashboards and cartographic products improving public service visibility.
- Improved forecast accuracy and reporting efficiency through statistical and automation techniques.

Education**Master's in Information Technology Management****Sep 2025**

- St Francis College, Brooklyn, New York

Bachelor of Technology – Computer Science Engineering**June 2023**

- Saveetha School of Engineering, Chennai, India

Certifications

- AWS Certified Solutions Architect – Associate (2024)
- Python Specialization – Coursera (2020)
- Google Data Analytics – Coursera
- AWS Security Fundamentals (2023)

Projects

- Procurement Optimization: Built Power BI dashboards to reduce procurement costs by 15% using trend analysis.
- Healthcare Analytics: Created a predictive diabetes detection model improving early diagnosis by 20%.
- Credit Card Fraud Detection: Developed ML-based fraud detection system achieving a 30% reduction in false positives.
- Emergency Response Mapping: Designed real-time mapping dashboards improving situational awareness for response teams.

- Stormwater Infrastructure Modernization: Digitization, QA/QC, and GIS integration for improved flood management.